Building Plan Approval

FRS Document

Submitted To:
State Urban Development Authority,
Govt. of Chattisgarh

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1 Scope

The scope of this document is to provide a detailed specification for the proposed solution for the ABPAS III application. Detailed information on the following aspects is available in this report

- Functional Scope
- User Access Level Specifications & Accountability Matrix
- Geographical Scope
- Non-Functional Requirement

1.1 Functional Scope

The functional scope is mentioned as follows –

- Processes and use cases

Processes & Use Cases

Process	Use Case
I. Registration & Login Process	1. Register Architects/Citizen 2. Approve/ Reject Registration by the CMO a. (Citizen registration does not require approval). 3. Login 4. Registration Renewal
II. User & Role Management Process	5. Create Role/User
III. Desktop Tool (offline)	6. Desktop Scrutiny Tool

IV.Submission,Approval, Clearance, Return & Rejection of building plan Process	7. Erect (New Plan), Alteration (Addition/Alteration), Addition/Alteration in Proposed Plan, Revalidation, Compounding & Compounding of Contrary (More construction from Approved Permission) plan submission by a Architect/Citizen 8. Demolition plan submission by a Architect/Citizen 9. Construction of Boundary Wall Plan submission by a Architect/Citizen 10. Clear or Return building plan (by the Building Clerk) 11. Clear or Return building plan (by the Building Inspector)Clear/Return/Approve/Reject building plan (by the Sub-Engineer/Assistant Engineer/Executive Engineer/Superintendent Engineer) 12. Clear/Return/Approve/Reject building plan (by CMO) 13. Colonizer Registration 14. Colony Development Permission 15. Application for Completion of Development Works Certificate
V. Creation/Verification of New Colony Data Process	17. Create/Verify (by the Colony Cell) the New Colony Data
VI. Resubmission Process	18. Erect (New Plan), Alteration (Addition/Alteration), Addition/Alteration in Proposed Plan, Revalidation, Compounding & Compounding of Contrary (More construction from Approved Permission) plan resubmission by a Architect/Citizen
VII. Demand Note and Sanction Letter Generation Process	19. Generation of Demand Note 20. Generation of Sanction Letter
VIII. Model Drawing Submission Process	21. Model drawing submission by the Architect/Citizen
IX. Track Building Plan Approval status by Owner	22. Track Building Plan Approval status by Owner
X. Citizen Services	23. Work Commencement Notice 24. Plinth level Inspection Certificate 25. Service Certificate 26. Completion cum Occupancy Certificate 27. Citizen Search 28. Make Payment Online
XI. Other	29. Access EODB Report 30. SLA Timeline Display Page & FAQ Section 31. Layout Sketches

1.1.1 Detailed Processes & Use Cases

Registration &Login Process

This process describes the Registration of Architects from various ULB's in Building plan approval system. Also, it describes the logging into the application for different users. The login process remains same for all authorized users. However, the information availability shall vary as per the access levels.

Consultan/Citizen Registration

Use Case 1.1: Register Architect

Description

- This use case describes the process of Registration of a Architect in BPAS for a particular ULB.

Actors

- Architect/Citizen
- CMO

Assumptions

- Architect is already empanelled with the particular ULB

Primary Business Scenarios

- Architect needs to type the application URL in a web browser
- Architect clicks the Sign-Up link on the home page of BPAS webpage
- The New User Registration page should appear
- The Architect enters the following details:
  - Personal Details - Name, User type, Qualification, Registration Number (If any), Experience (If Any)
  - Contact Details - Address, Mobile, Email, State, City, & Pin code
  - Login Details - User ID, Password, & Confirm password
  - Choose ULB (Urban local Body) - Select any ULB out of 413 ULBs.
  - Upload Document – Passport size Photo, Degree & ID proof (Licence, voter ID, Pan no & Samagra ID)
- After filling all the details Architect clicks on Submit button.
- and 'Approves' the Login Credentials of the Architect.
- The CMO verifies the details, if finds any incorrect details or documents, or if additional documents are required, the CMO can:
  - Return the registration to the Architect for corrections or additional documentation.
  - Reject the registration based on the findings.
- If the registration is returned, the Architect provides the required details or uploads the correct documents and resubmits the registration.
- The CMO re-evaluates the updated submission:
  - If everything is correct, the CMO generates a Demand Note/Payment Request.
- The Architect makes the required payment.
- Upon receipt of the payment in the ULB account, the CMO approves the registration/Login Credentials.
- The qualifications and documents required for each designation are as per BVN 2012 Rule 26(2).

Post-conditions

- The User will be able to login after the approval of the CMO.

Citizen Registration Process

This process describes the registration of citizens in the ABPAS-3.0 system. The system allows only one type of citizen registration, which is straightforward and does not require any approval or renewal.

Use Case 1.2: Citizen Registration

Description: This use case describes the process of registering a citizen in ABPAS-3.0 for a particular ULB.

Actors:

- Citizen

Assumptions:

- Citizens are allowed to register without the need for any approval.
- There is no need for renewal once the citizen is registered.
- Primary Business Scenarios:
- Citizen accesses the ABPAS-3.0 portal using a web browser.
- Citizen clicks the Sign-Up link on the homepage of the ABPAS-3.0 webpage.
- The New User Registration page appears.
- The citizen selects the appropriate radio button in the pop-up window to choose the user type.
- The citizen selects "Citizen" as the user type from the available options: Architect or Citizen.
- The citizen enters the following details:
  - Personal Details - Name, User type, Qualification, Registration Number (If any), Experience (If Any)
  - Contact Details - Address, Mobile, Email, State, City, & Pin code
  - Login Details - User ID, Password, & Confirm password
  - Choose ULB (Urban local Body) - Select any ULB out of 413 ULBs.
  - Upload Document – Passport size Photo, Degree & ID proof (Licence, voter ID, Pan no & Samagra ID)
- Once all documents are uploaded, the citizen must check the declaration boxes according to applicable norms.
- The citizen clicks the Submit button.
- After submission, the system activates the citizen user account immediately:
  - A confirmation pop-up window appears.
  - A confirmation message is sent to the citizen's mobile number and email address.

Post-conditions:

- The citizen is immediately active in the ABPAS-3.0 system upon successful registration and can log in without any need for approval or renewal.

Use Case 02: Approve/Reject Architect Registration by the CMO

Description

- This use case describes the working of the CMO **approving/rejecting** the Architect application for registration in the respective ULB.

Actors

- CMO.

Pre-conditions

- The registration application is successfully submitted by the Architect.

Primary Business Scenarios

- CMO checks the Registration Application.
  - In the Dashboard Listed items are shown below,
    - Name
    - Registration No
    - Address
    - Mobile
    - Email
  - CMO verifies the Registration number and other documents submitted by the Architect.
  - CMO approves, returns and rejects the file.

Post-conditions

- In case of approval, Architect will become the authorized User.
- In case of rejection, file goes to Architect stating the reason for the rejection.

Use Case 03: Login

Description

- This use case describes the process of logging in of various authorized users on BPAS.

Actors

- All authorized users.

Pre-Conditions

- The user should have approved login credentials in BPAS.

Primary Business Scenarios

- Successful login
  - User needs to type the application URL in a web browser
  - The application login screen should appear.
  - The user should enter login credentials, Captcha and click on 'submit'.
  - The user will be redirected to the 'Dashboard' screen upon successful login.

Alternate Business Scenarios

- User forgot password
  - User needs to type the application URL in a web browser.
  - The application login screen should appear.
  - The user should click on the 'Forgot password' hyperlink.
  - User needs to enter his/ her registered email id and click 'Submit' in the 'Forgot password' screen.
  - The new password will be sent to the email id registered by the database.
- User forgot username
  - User needs to type the application URL in a web browser.
  - The application login screen should appear.
  - The user should click on the 'Forgot username' hyperlink.
  - User needs to enter his/ her registered email id and click 'Submit' in the 'Forgot username' screen.

- The existing username will be sent to the email id registered by the database.

- User types wrong username/ password
  - User needs to type the application URL in a web browser.
  - The application login screen should appear.
  - The user enters wrong login credentials and clicks 'Submit'.
  - Error message appeared for invalid username credentials.
  - If User enters invalid credentials for 3 times, then User Account will be blocked.
- User enters correct credentials but is denied login due to inactivity.
  - User needs to type the application URL in a web browser.
  - The application login screen should appear.
  - The user enters login credentials and click 'Submit'.
  - The user is denied login since the user has not been logged in for the past 3 months.

Post-conditions

- The User will be redirected to the 'Dashboard 'page.

Use Case 04: Renew Architect Registration

Description: This use case describes the process of renewing a Architect's registration in BPAS for a particular ULB.

Actors:

- Architect
- CMO (Chief Municipal Officer)

Assumptions:

- The Architect is already registered and the registration is nearing expiry or has expired.

Primary Business Scenarios:

- An alert message pops up on the Architect's console about the registration renewal.
- The Architect clicks on the Renewal tab in the Sign-Up Pop-Up window.
- The Architect enters their User ID and clicks on Generate OTP.
- The Architect enters the received OTP to proceed.
- A new window opens displaying prefilled details of the Architect.
- The Architect can modify their Experience Details and COA Registration (only for architects) and upload any necessary documents.
- The Architect submits the renewal form and makes the payment through the payment gateway.
- The system forwards the renewal request to the CMO for approval.
- The CMO reviews the updated details and documents:
  - If everything is correct, the CMO approves the renewal.
  - If additional documents or corrections are needed, the CMO returns the renewal request to the Architect for further updates

.

Post-conditions:

- The Architect's registration is renewed upon approval by the CMO, allowing continued access to the BPAS system.

II - User & Role Management Process

This process describes the various other functions of System Administrator like creating, deleting, blocking, unblocking, changing the Business Role, Users, Groups,

Use Case 05: Create Role/User

- This use case describes the process of creation of user and role (all system Users other than Architect) by the System Administrator of BPAS.

Actors

- System Administrator

Pre-conditions

- The 'System Administrator' (SA) profile / user will be created when the application is installed.
- User should be successfully logged in the BPAS.

Primary Business Scenarios

Granting user access

- SA shall see the 'User information' tab by default.
- SA shall enter all relevant details for User creation such as
  - First name
  - Middle name
  - Last name
  - Mobile number
  - Email
  - Postal address
  - Department
  - Designation
- Click 'Save'.
- SA moves to the 'Business Roles' tab. User creates the Roles as mentioned in the Roles Master Annexure.
- SA should be able to set reliability indicator against each indicator.
- Click 'Log out'.

Alternative Business Scenarios

- Deleting user
  - Administrator can delete any information (except mandatory information) for any user after receiving appropriate approvals from SUDA.
  - Administrator can soft delete system users as well.
  - No hard delete shall be permitted
- Blocking user
  - Administrator can block any user from accessing the ABPAS application after receiving appropriate approvals from SUDA.
- Unblocking user
  - Administrator can unblock any user for accessing the ABPAS application after receiving appropriate approvals from SUDA.
- Post-conditions
  - The Role is created.

III -Desktop Tool (offline)

Use Case 06: Desktop Scrutiny Tool

- This use case describes the process of Using this offline tool user can create new project or can also use already created project. After creating the project, the user should get the drawing in dwg format and the scrutiny report which can be used as a zip file while submitting the application and data should be in encrypted form
- Actors
  - Architect/Citizen

Pre-conditions

- The Architect must be approved by ULB.
- User can avail the Desktop Scrutiny Tool from there login home page of BPAS

Primary Business Scenarios

- Architect/Citizen fills the form on the web application and after submitting the drawing an encrypted file is generated.
- Architect/Citizen submits this encrypted file and Upload Drawing to the desktop application and gets the details.
- After precheck is completed and other parameters are checked in the desktop application Scrutiny and PDF drawing is generated.
- This scrutiny and PDF drawing used for application submission is also in encrypted form.

- Post-conditions
  - Once the user clicks on the Desktop Scrutiny Tool tab the tool will be easily downloaded and installed in the system. User can be accessed offline.

IV- Submission, Approval, Clearance, Return & Rejection of Building plan process

Use Case 07: Erect (New Plan), Alteration (Addition/Alteration), Addition/Alteration in Proposed Plan, Revalidation, Compounding & compounding of Contrary / More construction from Approved Permission plan submission by a Architect.

Description

- This use case describes the process of Erect (New Plan)/Re-erect (Construction after demolition)/Alteration plan (Addition & Alteration) submission by a Architect application for building plan approval.

Actors

- Architect/Citizen

Pre-conditions

- The user has been tagged as 'Architect' after CMO's approval.
- The user should successfully log in to BPAS.
- Assessment number and property tax number must be validated without any due taxes for the property." (This is to ensure all property-related dues are cleared before proceeding with the application.)

Primary Business Scenarios

- After Successful Login, Architect have to fill the following details for the Proposal:
- Case Type
  - Erect (New Plan)
  - Alteration (Addition/Alteration)

- Addition/Alteration in Proposed Plan
- Revalidation
- Compounding of Contrary / More construction from Approved Permission
- Compounding

- Type of plot: -

    - Single Plot
    - Multi Plotted Development

- If Architect selects the Alteration (Addition/Alteration), Addition/Alteration in Proposed Plan, Revalidation & Compounding of Contrary / More construction from Approved Permission: -
    - Previous Sanction No.
    - Previous Sanction Date
    - Is this Compounding Case -Y/N

- ARCHITECT INFORMATION - Details will be auto-populated
  - Type of Architect
  - Name
  - Mobile Number
  - Email id
- OWNERS INFORMATION - Architect fills the correct owner details
  - Name
  - Mobile Number
  - Address
  - Email id
  - Google Maps integration for capturing the exact plot location.
- DEVELOPERS INFORMATION
  - Building is use for
    - self use
    - selling
- LAOUT INFORMTION- I
  - Property ID
  - T&CP Permission No
  - Layout Approval Type
  - Special Area
  - Name of Road
  - select place
  - Change place

    - Users can change plot number or select plot via ULB > Zone > Ward > Colony > Select Plot.
    - If plot details are unavailable, users click “+” to add new plot details, which will be verified by Colony Cell Verification Officer before final approval.

    Note:

    - If the user selects the approved layout, then user should enter “Application ID Number of T&CP Layout” option should be available in the ABPAS- III system.
    - If the user selects other options instead of the Approved Layout, then "T&CP Permission Number" must be entered.

- If T&CP layout is available in the system then plot related fields should be auto filled.
- Approved Layout Type: ULB/T&CP (All the plots which are in central area (prior to the T&CP Act 1973) T & CP approved layout is not mandatory).

- LAOUT INFORMTION- II

Include details for "Layout Information - II" (like sensitive zones, Nazul area, etc.), "Plot Abutting Details" (Front, Rear, Side1, Side2), and additional plot-specific conditions.

- Is this Plot in Sensitive Zone- Y/N
- Is this Plot in Nazul Area - Y/N
- Is This Plot Qualifies for Additional FAR - Y/N
- Is this Plot Mortgaged - Y/N
- Is Additional Parking Check Required - Y/N
- Is EWS/LIG Sanctioned on this plot - Y/N
- Is This Plot in vicinity - Y/N
- Is This Type Design - Y/N
- Is This a Mixed Use- Y/N
- Is this Plot in Heritage Zone - Y/N
- Is this Group Housing Case - Y/N
- Is this Housing Board Case - Y/N
- Is this Government Building - Y/N
- Is This Plot on Commercial road- Y/N
- Is This a low-Density Residential Area- Y/N
- Is this a Slum Rehabilitation- Y/N
- Is this ABD Area- Y/N
- Simhastha Area- Y/N

- PLOT ABUTTING DETAILS
  - Corner plot - Y/N

Option-

- Front
- Rear
- Side1
- Side 2

Sub option -

- Road (Width in Meters)
- Park
- Plot
- Others

- ROAD DETAILS
  - Water drainage sanction - Y/N
  - Water excavation details - (Metal Road/Cement/Concrete/etc.)
- COST ESTIMATE
  - Total Estimated Cost (INR)

- Connected Load of Building
  - Climate Classification
  - Building Type- CompositeClimate/Hotdry (Select Sub Use from dropdown)
  - Building Classification- CoreandShell/Mixeduse/SingleOwned (Select Sub Use from dropdown)
  - Total Building Area- Assembly/Bussiness/HealthCare/Hospitality/Others (Select Sub Use from dropdown)
  - Total Conditional Area of Building
  - Total Unconditional Area of Building
  - No. of Floors for Commercial Building
  - Buildup Area
- GENERAL INFORMATION
  - Means of Access - One Side/Two Sides
  - Means of Access Length
  - Means of Access Width
  - Provision for Divyang
- PLOT INFORMATION

– Land Use	– Select Sub Use from dropdown
– Connected Contract Demand	– below 120 kVA/Above 120 kVA
– Land Sub Use	– Select Sub Use from dropdown
– Is Plot Irregular?	– Y/N
– Plot Width (M.)	– To be entered
– Plot Depth (M.)	– To be entered
– Gross Plot Area (sq.m.)	– To be calculated (Plot Width (M.) * Plot Depth (M.))

- BUILDING INFORMATION

– Building Use	– Select Sub Use from dropdown
– Connected Load of Building	– Upto 100 KW/Above 100 KW
– Building Activity	– Select Sub Use from dropdown
– Type of Building	– Select Sub Use from dropdown
– Type of Development	– Select Sub Use from dropdown
– Proposed Built up Area	– To be entered
– Proposed Ground Coverage (%)	– To be entered
– Proposed Building Height (M.)	– To be entered
– Presence of Lift/Escalator/MovingWalk	– Y/N
– Proposed No. of Floors	– To be entered
– Existing Built-Up Area	– To be entered
– Existing Ground Coverage	– To be entered
– Existing Building Height	– To be entered
– Existing No.of. Floors	– To be entered
– Front CompoundWall Height	– To be entered
– Existing FAR Area	– To be entered

- Type of Construction - Select Sub Use from dropdown

- Upload CAD Drawing
  - The Architect should upload a zip file of the Drawing which is being checked by the offline tool and is in encrypted form.
- Check List

– Name of Parameter	– Answer
– Does Plot colony exist?	– No/Yes
– Colony type if exists	– Illegal – Legal – Unauthorised Colonies
– In Case of legal colony, Plot Number	– To be entered
– Plot Abutting Details Front	– To be entered
– Plot Abutting Details Rear	– To be entered
– Plot Abutting Details Side Left	– To be entered
– Plot Abutting Details Side Right	– To be entered

- List of Mandatory documents (To be uploaded)
  - Self attested copy of the sale deed, lease deed (or) any other Document showing proof of ownership (or) of any legal right in the relation to the site.
  - Key plan
  - Supervision Certificate
  - Affidavit
  - Georeferencing information of the plot/land should upload in .shx (shape) file format.

- List of Conditional Mandatory documents (To be Optional Uploaded)
  - Approved Layout plan from competent authority
  - Earthquake proof certificate (By Architect)
  - Structure Design certificate (For High rise)
  - NOC from Airport Authority
  - NOC of EIA/STP from CG pollution control board
  - NOC from High rise Committee
  - NOC from all the concerned departments described in the list of approved development permission layout of T&CP department
  - Letter of authorization (From owner to Architect)
  - Fire NOC
  - Archaeological NOC
  - Nazul NOC
  - Lift for Furnishing Details Related to Installation of Lift, Escalator or Moving Walk

- **Declaration Form (Rule 16 – BVN 2012)** regarding the checklist for the Site Plan to be filled by the Architect *(Kindly refer Annexure – III)*

- Architect confirms the Disclaimer by clicking on the checkbox before submitting the application
  - I undertake and affirm that the document submitted electronically by me on behalf of my client are checked and verified by me and are replicating original document to the best of my knowledge and belief. I am aware that any manipulation from original document may lead to necessary legal action under IPC.

- Architect is directed for making payment.
  - Architect pays the required fees via Payment Gateway.
  - After successful payment, receipt is generated.
- Architect successfully submits the application.

Post-conditions

- After submitting the file unique Number is generated for Future Reference.
- The User will be redirected to the 'Dashboard 'page, for further actions or to view the application status.

Use Case 08: Demolition plan submission by an Architect

Description

- This use case describes the process of Demolition Plan submission by a Architect/citizen for approval from Urban Local Body Authority.

Actors

- Citizen
- Architect
- Building Clerk
- Building Inspector
- Final Approving Authority

Pre-conditions

- The user has been tagged as 'Architect' after CMO's approval.
- The user should successfully log in to BPAS.

Primary Business Scenarios

- Citizen as well as a Architect can select the option of Demolition
- (In case of an Architect) After Successful Login, Architect have to fill the following details for the Proposal:
- Applicant Type: Architect/Supervisor/Structural Engineer will be auto-populated with Name, Email-Id & Mobile No. basis on the Login Credentials
- Case Type
  - Demolition
- Architect selects the Demolition:
- (Both Architect & Citizen have to enter the details) User enters the details in the following fields:
  - Did you obtain Building Permission Online: Y/N
    - If Yes then User is asked to enter the following details:
      - Building Permission No.
      - Then User clicks on Get Details Button & all the details including the previous sanction letter, owner details, approved drawing, etc. are auto-populated.

- User uploads the following documents:
  - NOC from all relevant Utilities Departments (Service provider) like water, electric, gas, sewer, and other service connections
  - Certificate from authorized technical personnel certifying the demolition
  - Photo of Site
- If No then User enters Building Permission No. and uploads the following documents:
  - Sanction Letter
  - Approved Drawing
  - Ownership related documents
  - NOC from all relevant Utilities Departments (Service provider) like water, electric, gas, sewer, and other service connections
  - Certificate from authorized technical personnel certifying the demolition
  - Photo of Site

- User fills the following Checklist: (in both Yes & No case)

1. Detailed study has been made of the structure to be demolished including its surroundings –Y/N
2. Necessary measures have been taken to prevent accidental collapse (hidden damages caused by fire, flood or earthquake) - Y/N
3. Prominent danger signs have been provided around the property - Y/N
4. All openings giving access to the structure have been barricaded and closed to all except the workmen - Y/N
5. During night warning lights have been placed on or above all barricades - Y/N
6. When work is not in progress, watchman has provided the information to the public to prevent unauthorized entry of the public into the danger zone - Y/N
7. All necessary safety appliances have been issued to all the workers before starting the work- Y/N
8. Safety distances to ensure the safety of the public have been clearly marked and indicated by signs - Y/N
9. Diversions for pedestrians have been ensured, where necessary for safety - Y/N

- User clicks on the Declaration checkbox:
  Declaration - I have carefully taken all the measures that are stated above in the checklist.

- Citizen/Architect successfully submits the application.

Note: Sample Sanction Letter is in Annexure IV

Use Case 09: Construction of Boundary Wall Plan submission by an Architect

Description

- This use case describes the process of Construction of Boundary Wall Plan submission by a Architect for approval/NOC from Urban Local Body Authority.

Actors

- Architect
- Building Clerk
- Building Inspector
- Final Approving Authority

Pre-conditions

- The user has been tagged as 'Architect' after CMO's approval.
- The user should successfully log in to BPAS.

Primary Business Scenarios

- After Successful Login, Architect have to fill the following details for the Proposal:
- Applicant Type: Architect/Supervisor/Structural Engineer will be auto-populated with Name, Email-Id & Mobile No. basis on the Login Credentials
- Case Type
  - Construction of Boundary Wall
- Architect selects the Construction of Boundary Wall:
- User enters the details in the following fields:
  - User enters the height of Boundary Wall

- User uploads the following documents:
  - Drawing of Boundary Wall
  - Ownership related documents
- Architect is directed for making payment.
  - Architect pays the required fees via Payment Gateway.
  - After successful payment, receipt is generated.

- Architect successfully submits the application.

Building Clerk & Building Inspector receives the alert on SMS, e-mail & dashboard. The Building Clerk verifies the documents as well the payment & the Building Inspector inspects the site. BC & BI presents a report to the Final Approving Authority. If the Architect has complied by the norms, Final Approving Authority issues the NOC for the construction of Boundary Wall. If there are any violation of the norms then Authority may return/reject the application.

NOTE: Sample NOC format is in Annexure IV

Use Case 10: Clear or Return Building plan (by the Building Clerk)

Description

- This use case describes the process of clearing/returning the Architect/Citizen application approval by the Building Clerk.

Actors

- Building Clerk.

Pre-conditions

- The user has been tagged as 'Building Clerk' by the System Administrator after obtaining appropriate approvals.
- The user should successfully log in to BPAS.
- The Citizen/Architect successfully submits the drawing and other mandatory documents.

Primary Business Scenarios

- Processing Pending Application

- In the Dashboard Listed items are shown below,
    a) Date of submission
    b) Reference No
    c) Architect Name
    d) Applicant Name
    e) Applicant Address
    f) Document Status (Pending, Cleared, InProcess or Returned)
    g) View Document (all the property related documents, NoC's, .shx (shape) file etc.)
    h) Action (Approve/ Return)
- The user selects any of the pending application.
- The user downloads all the property related documents, NoC's etc. and verifies them.
- The user clears or returns the file on the basis of verification.
- The user registers specific reasons/remarks for clearing as well as returning the application.

Alternate Business Scenarios

- Processing Re-submitted Application
  - The user selects any of the pending application which is re-submitted in the system.
  - The user downloads the documents and again verifies them.
  - The user clears or returns the file on the basis of verification.

Post-conditions

- The User will be redirected to the 'Dashboard 'page.
- On clearance the file goes to Level 1 officer of the ULB.
- On Return the file goes back to the Architect/Citizen.

Use Case 11: Clear or Return Building Plan (by the Building Inspector)

Description

- This use case describes the process of clearing/returning the Architect/Citizen application for building plan approval by the Building Inspector.

Actors

- Building Inspector

Pre-conditions

- The user has been tagged as 'Building Inspector' by the System Administrator after obtaining appropriate approvals.
- User should be successfully logged in the BPAS.
- The Citizen/Architect successfully submits the drawing and other mandatory documents.

Primary Business Scenarios

- Inspecting New Building Permission
  - In the Dashboard Listed items are shown below,
    a) Date of submission
    b) Reference No

c) Architect Name
d) Applicant Name
e) Applicant Address
f) Downloads (drawing in the format of PDF and .DWG)
g) Comments
h) Action (Approve/ Return)

- The user selects any of the pending application.
- The user downloads the drawings & goes for the Site Inspection & fills the following checklist:
- List of Checklist for Erection/Re-erection/Alteration of building
  (Kindly refer Annexure – III)
- The user clears or returns the file on the basis of verification.
- The user registers specific reasons for clearing as well as returning the application.

Post-conditions

- The User will be redirected to the 'Dashboard 'page.
- On clearance the file goes to Level 1 officer of the ULB.
- On Return the file goes back to the Architect/Citizen.

Use Case 12: Clear/Return/Approve/Reject Building Plan (by Building Officer)

Description

- This use case describes the working of the Building Officer **clearing/returning** the building permission application for building plan approval. The rights of **approving/rejecting** will only come once the Municipal Commissioner or CMO of the respective ULB, appoints the official as Building Officer as per Rule 6 of BVN 2012 for certain cases.

Actors

- Building Officer
- Sub-Engineer/Assistant Engineer/Executive Engineer/Superintendent Engineer as per the hierarchy in the ULB. For levels refer section 8.2 (User Access Level Specification & Accountability)

Pre-conditions

- The application is cleared by the Colony Cell or the Colony Data exists in the System.
- The application is successfully cleared by the Building Clerk, Building Inspector & the fully complaint scrutiny report by the Scrutiny Engine.

Primary Business Scenarios

- Sub-Engineer/Assistant Engineer/Executive Engineer/Superintendent Engineer checks the file
  - In the Dashboard Listed items are shown below,
    a) Date of Submission
    b) Reference No
    c) Architect Name
    d) Applicant Name
    e) Applicant Address
    f) BC Status
    g) BI Status

h) DCR Cell Status
i) Final Status
j) Downloads (Scrutiny Report and Drawing)
k) Action

- Sub-Engineer/Assistant Engineer/Executive Engineer/Superintendent Engineer downloads the Scrutiny Report, Drawing & other property documents as well as NoC's required for the verification.
- Sub-Engineer/Executive Engineer downloads the Scrutiny Report, Drawing & other property documents as well as NoC's (whichever) required for the verification.
- Sub-Engineer/Executive Engineer clears or returns the file as per his/her findings.

Alternative Business Scenarios

- Sub-Engineer/Assistant Engineer/Executive Engineer/Superintendent Engineer is appointed as the Building Officer by the Commissioner/CMO of the respective ULB.
- In the Dashboard Listed items are shown below,
  i. Date of Submission
  ii. Reference No
  iii. Architect Name
  iv. Applicant Name
  v. Applicant Address
  vi. BC Status
  vii. BI Status
  viii. DCR Cell Status
  ix. Final Status
  x. Downloads (Scrutiny Report and Drawing)
  xi. Action
- Building Officer approves or rejects the file as per his/her findings.
- In case of approval, the Building Officer issues the Sanction Letter.

Post-conditions

- In case of approval, the Building Officer issues the Sanction Letter.
- In case of clearance, file goes to CMO for the final approval.
- In case of return, file goes to Architect/BC/BI/ (L1 to L2) stating the reason for the return.

Use Case 13: Clear/Return/Approve/Reject Building Plan (by CMO)

Description

- This use case describes the working of the CMO/Commissioner **Approve/Reject/Return** the Citizen/Architect application for building plan approval & issuing the Sanctioning letter in case of approval.

Actors

- CMO/Commissioner

Pre-conditions

- The application is cleared by the Colony Cell or the Colony Data exists in the System.
- The application is successfully cleared by the Building Clerk, Building Inspector, Scrutiny Engine & Sub-Engineer/Assistant Engineer/Executive Engineer/Superintendent Engineer (L1 to L2).

Primary Business Scenarios

- CMO/Commissioner checks the file.
  - In the Dashboard Listed items are shown below,
    a) Date of Submission
    b) Reference No
    c) Architect Name
    d) Applicant Name
    e) Applicant Address
    f) BC Status
    g) BI Status
    h) DCR Cell Status
    i) L1 to L2 Status
    j) Final Status
    k) Downloads (Scrutiny Report and Drawing)
    l) Action
  - CMO/Commissioner downloads the Scrutiny Report, Drawing & other property documents as well as NoC's required for the verification.
  - CMO/Commissioner approves returns or rejects the file as per his/her findings.
  - In case of approval, Sanction Letter is digitally signed by the CMO/Commissioner.

Post-conditions

- In case of rejection, file goes to Architect stating the reason for the rejection.
- In case of return, file goes to Architect/L1-L2 stating the reason for return.
- CMO/Commissioner is the final authority for approving or rejecting the building plan permission in the ULB.
- If the Architect does not submit a proper reply within 7 days, the file will be automatically rejected.

Note: The final sanction letter format is displayed in Annexure-IV.

Colonizer Registration

Use Case 14: Colonizer Registration

Description

- This use case describes the process of registering an individual or organization as a colonizer in the Applicant Registration system.

Actors

- Colonizer/Applicant
- Approving Authority

Assumptions

- The applicant is either an individual or a firm/company seeking registration.

Primary Business Scenarios

- The applicant needs to access the registration system via a web browser.

- The applicant clicks the "Sign Up" link on the home page of the Colonizer Registration webpage.
- The New User Registration page should appear.
- The applicant enters the following details:
    - Personal Details - Name, Father's Name, Aadhar No., PAN No., Occupation
    - Organization Details (If applicable) - For firms/companies, details of all partners/directors such as Name, Address,
    - Nationality, PAN No., Designation
    - Contact Details - Current Address, Permanent Address, Mobile, Email
    - Login Details - User ID, Password, Confirm Password
    - Upload Documents - PAN Card, Company Registration, Previous Land Development Proof, Affidavit
- After filling in all the details, the applicant clicks on the "Submit" button.
- The Concern Authority verifies the application and documents; if discrepancies are found, the application can be: – Returned to the applicant with remarks for corrections.
- Rejected based on findings.
- If returned, the applicant corrects the details or uploads the required documents and resubmits the application.
- The Concern Authority re-evaluates the updated submission: – If everything is correct, the application is forwarded to the Approving Authority.
- The Approving Authority reviews and either: – Issues a Demand Note/Payment Request, or
  Returns the application for corrections.
- The applicant pays the registration fee of Rs 50,000 online.
- Upon receipt of payment, the Approving Authority generates and digitally signs the Registration Certificate, valid for 5 years, which the applicant can download.
- Notifications regarding registration status, expiry, and renewal options are sent via SMS and email.
- For renewal, the applicant can apply 30 days before the expiry date and pay a fee of Rs 25,000.

Post-conditions

- The applicant will be able to download the Registration Certificate after the approval of the Approving Authority.
- The applicant can track the status of the application through their console.
- Notifications and alerts are sent to the applicant at every stage of the process.

Note: The final Registration certificate format is displayed in Annexure- II.

Conlony Development Permission Application

Use Case 15: Colony Development Permission

Description

- This use case describes the process of applying for development permission, including the submission of necessary documents, fee payment, status tracking, and issuance and renewal of the Development Permission Certificate.

Actors

- Colonizer/Applicant
- Sub Engineer
- Officer in Charge (OiC)
- Approving Authority

Assumptions

- The applicant is either an individual or a firm/company seeking development permission.

Primary Business Scenarios

- The applicant logs in using registered credentials on the development permission system.
- The applicant clicks the "New Plan Submission" link on the Colonizer Dashboard.
- The New Development Permission Application page should appear.

- The applicant enters the following details:
  - **Personal/Company Details** - Auto-filled information such as Name, Father's Name, Email ID, Mobile No., PAN No., Aadhaar No., Occupation, Nationality, Current and Permanent Address.
  - **Bank/Banker Information** - Bank/Banker Name, Address.
  - **Architects/Contractors Information** - Architect, Structure Engineer, Contractor, Others (with fields for Name and Address).
  - **Layout Information** - ULB Name, Zone, Ward, Colony Name, Number of Plots, Number of Parking Areas in Colony, Available Open Areas in Colony, Estimated Cost of Internal Development.
  - **Checklist** - Answers to whether the applicant has developed any colony/land before the application, and whether the applicant is convicted of any offense described in Rule.
  - **Document Upload** - Required documents such as Registration Certificate, PAN card, Aadhaar card, proposed land details, affidavits, RERA registration details, and others.
- After filling in all the details, the applicant clicks the "Submit" button.
- The Sub Engineer verifies the application and documents, and if satisfactory, forwards it to the Development Permission (D.P.) Committee.
- The OiC forwards the application to the Approving Authority.
- The Approving Authority reviews and either: –
  - Issues a Demand Note/Payment Request, or
  - Returns the application for corrections.
- The applicant pays the development permission fee online, calculated as 0.50% of the Collector Guideline rate of diverted land.
- Upon receipt of payment, the Approving Authority generates and digitally signs the Development Permission Certificate, valid for 5 years.
- The Development Permission Certificate is made available for download in the applicant's console.
- Notifications regarding application status, certificate expiry, and renewal options are sent via SMS and email.
- The applicant can apply for renewal 30 days before the expiry date, with an option for revalidation available for 1 year.

Post-conditions

- The applicant will be able to download the Development Permission Certificate after the approval of the Approving Authority.
- The applicant can track the status of the application through their console.
- Notifications and alerts are sent to the applicant at every stage of the process.
- A list of all colonies approved for development is displayed on the home page.

Completion of Development Works Application

Use Case 16: Application for Completion of Development Works Certificate

Description

- This use case describes the process from the perspective of the applicant (colonizer) for obtaining the Completion of Development Works Certificate, including the submission of required documents, tracking the status of the application, and receiving the final certificate.

Actors

- Colonizer/Applicant
- Technical Officer
- Approving Authority

Assumptions

- The applicant has completed the development works and is ready to apply for the completion certificate.

Primary Business Scenarios

- The applicant logs in using registered credentials on the completion certificate system.

- The applicant clicks the "Completion Certificate" link on the Colonizer Dashboard.
- The Completion Certificate Application page should appear.
- The applicant enters the following details:
  - Colonizer/Developer Information - Inputs for Nagar Palika Nigam, District, Village, Khasra No., Applicant Name, Address, Email ID, Mobile No., Colonizer Registration Number, Firm/Company Name.
  - Infrastructure/Development Details - Details related to various infrastructure elements such as road construction, culvert, drainage, water supply, sewage line, solid waste management, electrification, development of open land, plantation, rainwater harvesting system, and other internal works.
- For each infrastructure element, the system requires:
  - Approval Development Permission: Pre-approved specifications (e.g., Bitumen Road, RCC culvert, Brick drainage).
  - Provision of Accomplished Development Work: User input for actual work completed.
  - Comments of Technical Officer: Inputs from the Technical Officer reviewing the work.
- After filling in all the details, the applicant clicks the "Submit" button.
  The application is then processed by the concerned authorities, with updates provided to the applicant at each stage.
  The Technical Officer reviews the accomplished development work and inputs comments on the system.
- Once the review is completed, the Approving Authority verifies the application and either:
  - Approves the application, or
  - Returns the application for resubmission with corrections.
- Upon final approval by the Approving Authority, the Completion Certificate is generated, signed digitally, and made available for download in the applicant's console.
- Notifications regarding application status, approval, and certificate issuance are sent via SMS and email.
- The colony is added to the Building Permission Portal once the certificate is issued.

Post-conditions

- The applicant will be able to download the Completion Certificate after approval.
- The applicant can track the status of the application through their console.
- Notifications and alerts are sent to the applicant at every stage of the process.
- The colony is listed on the Building Permission Portal upon completion.

Note: The Completion of Development Works Certificate format is displayed in Annexure- II.

V- Creation/Verification of New Colony Data process

Use Case 17: Create/Verify (by the Colony Cell) the New Colony Data

Description

- This use case describes the process of clearing/returning the Citizen/Architect application for building plan approval by the Colony Cell.

Actors

- Citizen/Architect
- Colony Cell Verification Officer/ Technical Officer
- CMO/Commissioner

Pre-conditions

- The user should successfully log in to BPAS.
- The Citizen/Architect successfully submits the drawing and other mandatory documents.
- The Colony data entered by Citizen/Architect which does not exists in the System.

Primary Business Scenarios

- Inspecting the New Colony
  - The user selects any of the pending application.
  - The user downloads all the property related documents, NoC's etc. and verifies them.
  - The user verifies the following details mentioned by the Architect to add new plot:
    - Colony Name
    - Plot No.
    - Plot Width (in Metres)
    - Plot Depth (in Metres)
    - Plot Area (Sq. Metres)
    - Type of Housing (Detached, Semi-detached, RowHouse, HighRise, MidRise)
    - Maximum Permissible Ground Coverage (%)
    - Maximum Permissible FAR
    - Front MOS (in Metres)
    - Rear MOS (in Metres)
    - Side 1 MOS (in Metres)
    - Side 2 MOS (in Metres)
    - Maximum Permissible Building Height (in Metres)
    - Road Width (in Metres)
    - Maximum Permissible Built-up Area (in Sq. Metres)
    - Arear under Road Widening (in Metres)
    - Net Plot Area (Excluding road widening area) (in Sq. Metres)
    - Land Use (Residential, Commercial Zone, Industrial Zone, Recreation, Public & Semi-Public, Special Purpose, Transportation, Public Utilities & Facilities, Water Bodies, Agriculture)
    - Sub Use (Options according to the selection in Land Use field)
    - Built-Up Area of Existing Structure (in Sq. Metres)
    - Frontage (in Metres)
    - Is Plot Irregular (Yes/No)
    - Is EWS/LIG sanctioned on this Plot? (Yes/No)
    - Is Plot Mortgaged? (Yes/No)
  - The user clears or rejects the file on the basis of verification.
  - The user registers specific reasons for clearing as well as returning the application.
  - File goes to CMO for the approval.
  - In case of approval the new colony data is created & saved in the database.

Post-conditions

- The User will be redirected to the 'Dashboard 'page.

VI - Resubmission process

Use Case 18: Building Plan Resubmission by a Architect/Citizen

Description

- This use case describes the process of Building plan resubmission by a Architect/Citizen application for building plan approval.

Actors

- Architect/Citizen.

Pre-conditions

- The Building Clerk/Building Inspector/DCR Cell/L1-L2 returns the application to the Architect.

Primary Business Scenarios

- After successfully Login, Architect uploads all required documents and drawings.
- In case of file returned by the DCR Cell, Upload the new Cad Drawing in the Format of .dwg changed as per the scrutiny report non-compliances
- Any of the following documents returned by the Building Clerk/Building Inspector/L1-L2 needs to be uploaded. The rest of the documents cannot be modified by the Architect
- List of Mandatory documents (To be uploaded)
    - Self attested copy of the sale deed, lease deed (or) any other Document showing proof of ownership (or) of any legal right in the relation to the site.
    - Key plan {See rule 16(3) of BVR 2012}
    - Form 16 (Supervision Certificate)
    - E-Affidavit
- List of Conditional Mandatory documents (To be Optional Uploaded)
    - Approved Layout plan from competent authority
    - Earthquake proof certificate (By Architect)
    - Structure Design certificate (For High rise)
    - NOC from Airport Authority
    - NOC of EIA/STP from CG pollution control board
    - NOC from High rise Committee
    - NOC from all the concerned departments described in the list of approved development permission layout of T&CP department
    - Letter of authorization (From owner to Architect)
    - Fire NOC
    - Archaeological NOC
    - Nazul NOC
    - Lift for Furnishing Details Related to Installation of Lift, Escalator or Moving Walk
- Check List needs to be resubmitted if Building Inspector returns the application sighting any discrepancy in the following checklist filled by the Architect during the time of submission.
- The list of checklists is the same that is used in “Use Case #6” at the time of submission and filing of the new plan by the Architect/citizen.

(Kindly refer Annexure –

III) Post-conditions

- After resubmission the file goes to the official who has returned.

VII- Demand Note and Sanction Letter Generation process

Use Case 19: Generation of Demand Note

Description

- This use case describes the process of auto-generation of the Demand Note after the approval of the building plan application by the CMO.

Actors

- CMO
- Building Officer
- Building Clerk

Pre-conditions

- The application is approved by the CMO.

Primary Business Scenarios

- Building Officer approves the Building Plan application & enters the collector guide line rate.
- CMO verify System generated Demand Note based on the standard building permission fee as per the BVN Rules and other ULB specific charges (if any) & generates the Demand Note.
- If the Architect/citizen finds that earlier they had paid some fees for the same plot and the Architect/citizen has proof documents for the same.
- So the Architect/citizen should have "Change Demand Note" tab and "Upload" option, by clicking on this tab the user can request to change the demand note and upload valid documents.
- There should be an option for the CMO console to also check this request and if found OK then update and regenerate the demand note.

  Note- Change Demand Note needs to be provided only 1 time.
- After demand note is generated Architect is directed for making payment.
  - Architect/Owner pays the required fees via Payment Gateway.
  - After successful payment, receipt is generated.

Post-conditions

- In case of payment done within the 15 days, application is forwarded to the CMO to generate the Sanction Letter and for e-Sign.
- In case of non-payment, within 15 days, file goes to the CMO, who gives another 15 days extension. If fee isn't paid within this period, the CMO can reject the file.

Use Case 20: Generation of Sanction Letter

Description

- This use case describes the process of generation of the Sanction Letter after the verification of the building permission fee payment by the Building Clerk and digital signature of the CMO.

Actors

- CMO

Pre-conditions

- The building permission fee payment is verified by the Building Clerk. Primary Business Scenarios
- After the Building Clerk verifies the payment of the building permission fee, Sanction Letter is generated by CMO & digitally signs the Sanction Letter.

Post-conditions

- Architect/Citizen can download the Sanction Letter by logging into the application or through citizen search option.

VIII - Model Drawing Submission process

Use Case 21: Model drawing submission by the Citizen

Description

- This use case describes the process of Model Drawing submission by the Citizen for building plan approval.

Actors

- Citizen.

Pre-conditions

- The Model Drawings are available in the BPAS system.

Primary Business Scenarios

- Citizen types the BPAS URL in the browser and Home Page appears.
- Citizen clicks the Model Drawing link on the Home Page and is redirected to the Model Drawing Search Page.
- Citizen enters the following search criteria:
    - ULB Name
    - Plot Area (Sq. Mt.)
    - Plot Depth (Mt.)
    - Plot Width (Mt.)
    - FAR
    - Road Width (Mt.)
    - Permissible Ground Coverage
- After entering the details in the search criteria, Citizen clicks the search button
- List of the applicable drawings is shown on the screen.
- Citizen can see & select any of the drawing available in the list.
- Citizen enters the following details:
- Owner Information
    - Name
    - Mobile Number
    - Address
    - Email id
- Layout Information
    - Zone

- Ward
- Colony
- Plot No
- Plot Zone: Central Area/Non Central Area
- Plot Area
- Actual Area

- Is the plot leased by Housing Board or Development authority? Yes/No
- NOC of Housing Board/Development Authority is required in case of Yes.
- Is full/part of plot constructed before, for which demolition is required in order to have new construction? Yes/No
- In case of Yes as an answer, Affidavit for dismantling the old structure must come under the mandatory document list. List of documents to be uploaded:
  - Ownership Document
  - Site Photographs
  - Zonal NOC/Property Tax Receipt
  - Undertaking / Affidavit by owner for scanned documents are original & verified
  - NOC of Housing Board/Development Authority (Conditional Mandatory)
  - Affidavit for dismantling the old structure (Conditional Mandatory)
- Citizen pays the corporation fee and karmakar fee online.
- A unique application number with final note is generated & sent to the Citizen via SMS and email upon submission of the plan.

Post-conditions

- Application Home Page will be displayed after the submission.

Note:

- The final note for Model Drawing format is displayed in Annexure-IV.

IX - Track Building Plan Approval status by Owner

Use Case 22: Track Building Plan Approval status by Owner

Description

- This use case describes the process of tracking of the application for building plan approval by the Owner.

Actors

- Owner.

Pre-conditions

- The application for the Building Plan Approval is applied by the Architect or the Owner himself in case of Model Drawing.

Primary Business Scenarios

- The Owner as well as Architect gets an alert on e-mail & sms about the Building Plan Approval application with Unique Application Number.
- Owner types the URL for the BPAS system on the browser. On the Home Page under the Tab of Citizen Services, the Owner selects the option to track the file using the Unique Application Number. The application is redirected to the Citizen Tracking Page.
- Owner enters the Unique Application Number and clicks the Generate OTP button. Then a text box and submit

button appears to enter the OTP.

- Owner receives the OTP on his registered mobile phone. Owner enters the OTP and clicks on the Submit button. Application is redirected to the Citizen Dashboard, where he/she can see the file flow and the current status of the file.

Post-conditions

- After tracking the file status, Owner logs off from the application.

X - Citizen Services

This process describes the Work Commencement Notice, Plinth Level Inspection, Service, Completion cum Occupancy options for the Owner, Search option for the Citizens and & Payment option for the Owners. All the certificate formats are provided in section 21.4 Annexure IV.

Use Case 23: Work Commencement Notice

Description

- This use case describes the process of Work Commencement Notice after the successful building plan approval by the ULB authority.

Actors

- Architect/Citizen
- Owner

Pre-conditions

- The work commencement date must be in the period of 1 year from the sanction date of building permission.
- The application for the Building Plan Approval applied by the Architect/Owner is successfully approved by the ULB Authority.

Primary Business Scenarios

- Architect/Owner types the URL for the BPAS system on the browser.
- In case of Architect: On the Home Page Architect logs in and selects the option to apply for the Work Commencement Notice. The application is redirected to the Work Commencement Notice Page.
- In case of Owner: On the Citizen Services page, Owner selects the option to apply for the Work Commencement Notice. The application is redirected to the Work Commencement Notice Page.
- User enters the details in the following fields:
  - Did you obtain Building Permission Online: Y/N
  - If Yes then User is asked to enter the following details:
    - Building Permission No.
  - Then User clicks on Get Details Button & following fields are auto-populated
    - Sanctioned On
    - Permission Valid Till Date (System Generated)

- Application Details
  - Case Type
  - File No.
  - Nature of Proposal
- Details of Building Permission
  - Architect Type: Architect/Structural Engineer/ Supervisor

- Architect Name
- Registration No.
- Mobile No.

- Sanction Details
  - Sanction Issued on Date
  - Drawing Plan
  - Scrutiny Report
  - Building Permission

- Plot Details:
  - Marginal Open Space F , R, S1, S2
  - Gross Plot Area (Sqm)
  - Area of Road Widening (Sqm)
  - Net Plot Area (Sqm)
- Building Details:
  - Building Name
  - Building Use
  - Building Sub-Use
  - No. of Floors
- Site Address:
  - Zone
  - Ward
  - Colony
  - Plot No.
- Owner Details:
  - Name
  - Email Id
  - Mobile No.
  - ID Proof Number (Except Aadhaar number)
- User uploads the following documents:
  - Site photo
  - Appendix E-1 (Form for Work Commencement Notice)
- User enters the Date of Commencement of Work & clicks the “Intimate Now” button.

- If No then User is asked to enter the following details:
  - Building Permission No.
  - Sanctioned On
  - Permission Valid Till Date (System Generated)
  - Details of Building Permission
    - Architect Type: Architect/Structural Engineer/ Supervisor
    - Architect Name
    - Registration No
    - Mobile No.

- Building Details:
  - Building Name
  - Building Use
  - Building Sub-Use
  - No. of Floors
- Site Address:
  - Zone
  - Ward
  - Colony
  - Plot No.
- Owner Details:
  - Name
  - Email Id
  - Mobile No.
  - ID Proof Number (Except Aadhaar number)
- User uploads the following documents:
  - Copy of Sanctioned Plan
  - Copy of Building Permission
  - Site photo
  - Appendix E-1 (Form for Work Commencement Notice)
- User enters the Date of Commencement of Work & clicks the "Intimate Now" button.

Note: If applicant applies for work commencement after 1 year from the sanction date of building permission, then system shall give a popup to the applicant to that he/she is not allowed for the same & needs to get his/her building permission revalidated.

Use Case 24: Plinth level Inspection Certificate

Description

- This use case describes the process of Plinth level Inspection Certificate after the successful building plan and Work Commencement Notice approval by the ULB authority.

Actors

- Architect/Citizen
- Owner
- Building Clerk
- Building Inspector
- Final Approving Authority

Pre-conditions

- The application for the Building Plan Approval applied by the Architect/Owner is approved by the ULB authority and notice for Work Commencement Notice is given by the Authority.

Primary Business Scenarios

- Architect/Owner types the URL for the BPAS system on the browser.

- In case of Architect: On the Home Page Architect logs in and selects the option to apply for the Plinth Level Inspection Certificate. The application is redirected to the Plinth Level Inspection Certificate Page.
- In case of Owner: On the Home Page Citizen logs in and selects the option to apply for the Plinth Level Inspection Certificate. The application is redirected to the Plinth Level Inspection Certificate Page.
- User enters the details in the following fields:

- Did you obtain Building Permission Online: Y/N
- If Yes then User is asked to enter the following details:
  - Building Permission- No.

    Then User clicks on Get Details Button
    - Sanctioned On
    - Permission Valid till Date (System Generated)

- Whether intimation for commencement of work was given: Y/N
- In case of No, applicant is redirected to the Work Commencement Notice page.

In case of Yes:

Whether it was given Online: Y/N

- If Yes then System will ask the User to fill the following data:
  - Date of Commencement (Work Start Date)
  - Required (Plinth Part / Full)
  - Plinth completion date
- If No then System will ask the User to fill the following data:

- Date of Commencement (Work Start Date)
- Name of Architect/ Structural Engineer/Supervisor
- Registration No.
- Required (Plinth Part / Full)
- Plinth completion date
- Following Fields will be auto populated in both Yes & No case
  - Application Details
    - Case Type
    - File Number
    - Nature of Proposal
  - Details of Building Permission
    - Name of Architect/Structural Engineer/ Supervisor
    - Registration No.
    - Mobile No.
  - Sanction Details
    - Drawing Plan
    - Scrutiny Report
    - Building Permission
  - Plot Details:
    - Marginal Open Space (in meters) – Front, Rear, Side 1, Side 2
    - Gross Plot Area (in square meters)
    - Area of Road Widening

        - Net Plot Area (in square meters)
    - Building Details:
        - Building Name
        - Building Use
        - Building Sub-Use
        - No. of Floors
    - Site Address:
        - Zone
        - Ward
        - Colony
        - Plot
    - Owner Details:
        - Name
        - Email Id
        - Mobile No.
        - ID Proof Number (Except Aadhaar number)
  - User uploads the following documents:
    - Photograph of constructed area showing setbacks on 4 sides
    - Appendix E-2 (Form for Notice for Inspection of Building at Plinth Level)
    - Fire NOC (as per NBC)
- User clicks the "Intimate Now" button.

- If No then User is asked to enter the following details:
  - Building Permission No.
  - Sanctioned On
  - Permission Valid till Date (System Generated)

  - Whether intimation for commencement of work was given: Y/N
  - In case of No, applicant is redirected to the Work Commencement Notice page.
  - In case of Yes, User is asked to enter following details:
    - whether it was given Online: Y/N
    - If Yes, User is asked to fill the following details
      - Date of Commencement (Work Start Date)
      - Required (Plinth Part / Full)
      - Plinth Completion Date
      - Is this High-Rise Case: Y/N
      - In case of Yes User is asked to enter the Building Height (m)
      - In case of No User can proceed further
    - If No, User is asked to fill the following details:
      - Date of Commencement (Work Start Date)
      - Name of Architect/Structural Engineer/Supervisor

- Registration No.
- Required (Plinth Part / Full)
- Plinth Completion Date
- Is this High-Rise Case: Y/N
- In case of Yes User is asked to enter the Building Height (m)
- In case of No User can proceed further

- Details of Building Permission
  - Architect Type: Architect/Structural Engineer/ Supervisor
  - Architect Name
  - Registration No
  - Mobile No.
- Building Details:
  - Building Name
  - Building Use
  - Building Sub-Use
  - No. of Floors
- Site Address:
  - Zone
  - Ward
  - Colony
  - Plot No.

- Owner Details:
  - Name
  - Email Id
  - Mobile No.
  - ID Proof Number (Except Aadhaar number)

- User uploads the following documents:

  - Copy of Sanctioned Plan
  - Copy of Building Permission
  - Photograph of constructed area showing setbacks on 4 sides
  - Appendix E-2 (Form for Notice for Inspection of Building at Plinth Level)
  - Fire NOC (as per NBC)

- User clicks the “Intimate Now” button.

- Building Clerk & Building Inspector receives the alert on SMS, e-mail & dashboard. The Building Clerk verifies the

documents & the Building Inspector inspects the site, fills the Plinth Inspection List (below) and both BC & BI presents a report to the Final Approving Authority. If the Architect has complied by the norms, Final Approving Authority issues the Note for Plinth Level Completion in cases other than high-rise. If there are any violation of the norms then Authority may hold the application. Plinth level certificate is issued only in case of High-Rise buildings

Note:

- The Certificate format is displayed in Annexure-IV.
- Applicant can view remarks and the application flow in the user interface.

Plinth Inspection List

- Site Visit Done
  Date:

1. Is Construction done as per approved drawing? Yes/No
   - Front (m)
   - Rear (m)
   - Side 1 (m)
   - Side 2 (m)
2. If no, construction done as per approved drawing –
   a. Is any violation done at constructed building, can it be rectified? If it can't be rectified then Municipal Engineer shall issue certificate after demolition of the violated area, or reconstruction Yes/No
      - Front (m)
      - Rear (m)
      - Side 1 (m)
      - Side 2 (m)

   a. If MOS violated is compoundable? If it is not compoundable then only after demolition municipal engineer shall issue the certificate Yes/No
      - Violations
      - Remarks

   b. If done Construction is for such a purpose which may affect health and convenience of nearby residents according to authority opinion Yes/No

   c. Is any other violation? Yes/No
      - Violations (m)
      - Remarks
3. If the Building constructed above the Municipal sewers, sewage drain or mains or under transmission line? Yes/No
4. Please verify if following facilities provided at site:

   a. Are proper toilets provided? Yes/No

   b. Is drinking water provision available? Yes/No

   c. Are relevant amenities given? Yes/No

NOTE:

- SLA for the plinth inspection & approval is 15 days for the high-rise buildings and 7 days for all other buildings. If no action is taken by the competent authority in the SLA, deemed permission is granted to the applicant in cases other than high-rise, stating that he can commence work as per approved plan.
- Reminder SMS must be sent to the Building Inspector for site inspection within 5 days if Building Inspector doesn't do site

inspection within 7 days, an SMS must be sent to Building Inspector that he has not visited the site & deemed approval is granted. If any violation comes up then it will be the responsibility of the Building Inspector.

Use Case 25: Service Certificate

Description

- This use case describes the process of Service Certificate after the successful building plan approval, Work Commencement Notice approval and Plinth level inspection by the ULB authority.

Actors

- Architect/Citizen
- Owner
- Building Clerk
- Building Inspector
- Final Approving Authority

Pre-conditions

- The application for the Building Plan Approval applied by the Architect/Owner, Work Commencement as well as Plinth Level Completion Certificate is approved by the ULB Authority.

Primary Business Scenarios

- Architect/Owner types the URL for the BPAS system on the browser.
- In case of Architect: On the Home Page Architect logs in and selects the option to apply for the Service Certificate. The application is redirected to the Service Certificate Page.
- In case of Owner: On the Home Page Citizen logs in and selects the option to apply for the Service Certificate. The application is redirected to the Service Certificate Page.
- User enters the details in the following fields:
  - Did you obtain Permission Online: Y/N
  - If Yes then User is asked to enter the following details:
    - Building Permission No.
    - Then User clicks on Get Details Button & following fields are auto-populated
      - Sanctioned On
      - Permission Valid till Date (System Generated)
      - Whether intimation for commencement of work was given: Y/N
        - In case of No, applicant is redirected to the Work Commencement Notice page
        - In case of Yes, applicant is asked about the following details:
          - Whether it was given online: Y/N
            - In case of Yes, applicant is asked to enter the following details:

              oDate of Commencement (Work Start Date)

              oWhether intimation for Plinth was given? Y/N
              - In case of No, applicant is redirected to the Intimation for Plinth Level Page
              - In case of Yes, applicant is asked the following details:
              - Plinth Completion Date
              - Required (Plinth Part / Full)
              - Service Date
            - Below details are auto populated:

oApplication Details

- Case Type
- File No.
- Nature of Proposal

oDetails of Building Permission:

- Architect Type: Architect/Structural Engineer/ Supervisor
- Architect Name
- Registration No
- Mobile No.

- Sanction Details
  - Drawing Plan
    - Scrutiny Report
    - Building Permission

oPlot Details:

- Marginal Open Space (in meters) – Front, Rear, Side 1, Side 2
- Gross Plot Area (in square meters)
- Area of Road Widening
- Net Plot Area (in square meters)

oBuilding Details:

- Building Name
- Building Use
- Building Sub-Use
- No. of Floors

oSite Address:

- Zone
- Ward
- Colony
- Plot

oOwner Details:

- Name
- Email Id
- Mobile No.
- ID Proof Number (Except Aadhaar number)

- User uploads the following documents:
  - Service Notice
  - Site photo
- User clicks the “Intimate Now” button.

- If case of No applicant is asked to enter the following details:
  - Building Permission No.
  - Sanctioned On
  - Permission Valid Till Date (System Generated)

- Whether intimation for commencement of work was given: Y/N
  - In case of No, applicant is redirected to the Work Commencement Notice page.
  - In case of Yes, applicant is asked to give following details:
    - Whether it was given online? Y/N
      - In case of Yes, the applicant enters the Date of Commencement
        - Whether intimation for plinth inspection was given: Y/N

          oIn case of No, applicant is redirected to the Plinth Inspection page.

          oIn case of Yes, applicant is asked the following details
            - Whether it was given online? Y/N
            - In case of Yes, the Architect enters the following details:
            - Plinth Completion Date
            - Required (Plinth Part / Full)
            - Service Date
            - Is this a high rise case? Y?N
            - In case of Yes, applicant enters the Building Height
            - In case of No, the Architect proceeds further & following details are auto populated

Application Details:

- Case Type
- File No.
- Nature of proposal Details of Building Permission

Architect Type: Architect/Structural Engineer/ Supervisor

- Architect Name
- Registration No
- Mobile No.

Building Details:

- Building Name
- Building Use
- Building Sub-Use
- No. of Floors
- Site Address:
- Zone
- Ward
- Colony
- Plot No.

Owner Details:

- Name
- Email Id
- Mobile No.
- ID Proof Number (Except Aadhaar number)

- User uploads the following documents:
  - Copy of Sanctioned Plan
  - Copy of Building Permission
  - Photograph of constructed area showing setbacks on 4 sides

- Form for Notice for issue of Service Certificate
- User clicks the "Intimate Now" button.

- Building Clerk & Building Inspector receives the alert on SMS, e-mail & dashboard. The Building Clerk verifies the documents & the Building Inspector inspects the site, fills the Service Check List and both BC & BI forwards the case to the Final Approving Authority for approval. If the Owner/Architect has complied by the norms, Final Approving Authority issues the certificate for Service. If there are any violation of the norms then Final Approving Authority may reject the application on the basis of non-compliance (deviations from the approved drawing etc.).

NOTE: If any previous forms aren't filled, system must redirect applicant to that form first. The Certificate format is displayed in Annexure-IV.

Applicant can view remarks and the application flow in the user interface.

Service Inspection List

File number: Architect: File received:

Site Visit done date:

1. Is construction done as per approved drawing? Y/N
2. Construction done as per approved drawing- If YES
   a. Is building material removed from site Y/N
   b. Is site clean, neatly maintained and no seepage or leakage in walls Y/N

Construction done as per approved drawing- If NO

a. Is any violation done at constructed building, can it be rectified? If it can't be rectified then municipal engineer shall issue certificate after demolition of violated area or reconstruction – Y/N
   *In case of yes, box will come up for Building inspector to list the MOS violations
   Front [ ] Rear [ ] Side1 [ ] Side2 [ ]
b. Is MOS violated at constructed site? Y/N
c. If MOS violated is compoundable? If it is not compoundable then only after demolition municipal engineer shall issue certificate. Y/N
d. If done construction is for such a purpose which may affect health and convenience of nearby residents according to authority opinion. Y/N

3. If the building constructed above municipal sewers, sewage drain or mains or under transmission line? Y/N
4. If the owner has explained all necessary measures to the authority for securing the construction work from unsocial elements? Y/N
5. Distance of nearby police station from constructed building? Y/N
6. If building use has not been clarified by the authority that it would be used for people gathering like cinema, theatre, general worship? Y/N
7. Is there any corporation tax pending on constructed building till date? Y/N

Use Case 26: Completion cum Occupancy Certificate

Description

- This use case describes the process of Completion cum Occupancy Certificate after the successful building plan approval, Work Commencement Notice, Plinth level Inspection by the ULB authority.

Actors

- Architect/Citizen
- Building Clerk
- Building Inspector
- Final Approving Authority

Pre-conditions

- The application for the Building Plan Approval applied by the Architect is approved by the ULB Authority.
- Work Commencement, Plinth Level Completion Certificate approved by the ULB Authority.

Primary Business Scenarios

- Architect/Owner types the URL for the BPAS system on the browser.
- In case of Architect: On the Home Page Architect logs in and selects the option to apply for the Completion cum Occupancy Certificate. The application is redirected to the Completion cum Occupancy Certificate Page.
- In case of Owner: On the Home Page Citizen logs in and selects the option to apply for the Service Certificate. The application is redirected to the Completion cum Occupancy Certificate Page.
- User enters the details in the following fields:
  - Did you obtain Building Permission Online: Y/N
  - If Yes then User is asked to enter the following details:
    - Building Permission No.
    - Then User clicks on Get Details Button & following fields are auto-populated
      - Sanctioned On
      - Permission Valid Till Date (System Generated)
      - Whether intimation for commencement of work was given: Y/N
        - In case of No, applicant is redirected to the Work Commencement Notice page.
        - In case of Yes, applicant is asked to provide the following details:
          - Whether it was given online? Y/N

- In case of No, applicant is asked enter the following details:
  oDate of Commencement (Work Strat date)
  oName of Architect/Structural Engineer/Supervisor
  oRegistration No.

- In case of Yes, applicant is asked enter the following detail:
  oDate of Commencement of Work

Applicant proceeds further (Same flow both in case of Yes or No)

- Whether intimation for Plinth was given? Y/N

oIn case of No, applicant is redirected to the Plinth Inspection page.

oIn case of Yes, the applicant is asked to enter the following details:

- Whether it was given online? Y/N

- In case of Yes, the applicant enters the Plinth Completion Date
- Required (Plinth Part / Full)

- In case of No, applicant is asked enter the following details:
- Date of Plinth Completion Date
- Name of Architect/Structural Engineer/Supervisor
- Registration No.

Applicant proceeds further (Same flow both in case of Yes or No)

- Whether intimation for Service was given? Y/N

- In case of No, the applicant is allowed to proceed further as Service Certificate is not mandatory.

- In case of Yes, the applicant is asked the following details:

- Whether it was given online? Y/N

- In case of Yes, the applicant is asked to enter the Service Date.

- In case of No, applicant is asked enter the following details:
- Service Date
- Name of Architect/Structural Engineer/Supervisor
- Registration No.

Following fields are auto populated:

- Case Type
- File No.
- Architect Type: Architect/Structural Engineer/ Supervisor
- Architect Name
- Registration No
- Mobile No.
- Drawing Plan
- Scrutiny Report
- Building Permission
- Plot Details:
  - Marginal Open Space (in meters) – Front, Rear, Side 1, Side 2
  - Gross Plot Area (in square meters)

        - Area of Road Widening
        - Net Plot Area (in square meters)
    - Building Details:
        - Building Name
        - Building Use
        - Building Sub-Use
        - No. of Floors

    The applicant fills the following fields:

Head	As per permission	Actual Construction
Total Built-up Area	System will auto-populate in case of online permission	
Area deductible for FAR	System will auto-populate in case of online permission	
Area counted in FAR	System will auto-populate in case of online permission	
Compounding Permission No.		

- Site Address:
    - Zone
    - Ward
    - Colony
        - Plot
    - Owner Details:
        - Name
        - Email Id
        - Mobile No.
        - ID Proof Number (Except Aadhaar number)
- User uploads the following documents:
    - Appendix E-3 (Form for Notice for Completion Permit)
    - Appendix E-4 (Form for Notice for Occupancy Permit)
    - Fire NOC (as per NBC)
    - Site photo
- Completion cum Occupancy date
- User clicks the “Intimate Now” button.

- If No then User is asked to enter the following details:
    - Building Permission No.
    - Sanctioned On
    - Permission Valid Till Date (System generated – 3 years post the Sanction Date)
    - Case Type
    - File No.

- Architect Type: Architect/Structural Engineer/ Supervisor
- Architect Name
- Registration No
- Mobile No.
- Building Details:
  - Building Name
  - Building Use
  - Building Sub-Use
  - No. of Floors
- The applicant fills the following fields:

Head	As per permission	Actual Construction
Total Built-up Area		
Area deductible for FAR		
Area counted in FAR		
Compounding Permission No.		

- Site Address:
  - Zone
  - Ward
  - Colony
  - Plot
- Owner Details:
  - Name
  - Email Id
  - Mobile No.
  - ID Proof Number (Except Aadhaar number)
  - Whether intimation for commencement of work was given: Y/N
    - In case of No, applicant is redirected to the Work Commencement Notice page.
    - In case of Yes, applicant is asked to provide the following details:
      - Whether it was given online? Y/N

        - In case of No, applicant is asked enter the following details:
          oDate of Commencement (Work Strat date)
          oName of Architect/Structural Engineer/Supervisor
          oRegistration No.
        - In case of Yes, applicant is asked enter the following detail:
          oDate of Commencement of Work

          Applicant proceeds further (Same flow both in case of Yes or No)

        - Whether intimation for Plinth was given? Y/N

oIn case of No, applicant is redirected to the Plinth Inspection page.

oIn case of Yes, the applicant is asked to enter the following details:

- Whether it was given online? Y/N

- In case of Yes, the applicant enters the Plinth Completion Date
- Required (Plinth Part / Full)

- In case of No, applicant is asked enter the following details:
- Date of Plinth Completion Date
- Name of Architect/Structural Engineer/Supervisor
- Registration No.

Applicant proceeds further (Same flow both in case of Yes or No)

- Whether intimation for Service was given? Y/N

- In case of No, the applicant is allowed to proceed further as Service Certificate is not mandatory.

- In case of Yes, the applicant is asked the following details:
- Whether it was given online? Y/N
- In case of Yes, the applicant is asked to enter the Service Date.
- In case of No, applicant is asked enter the following details:
- Service Date
- Name of Architect/Structural Engineer/Supervisor
- Registration No.

- User uploads the following documents:
  - Copy of Sanctioned Plan
  - Copy of Building Permission
  - Photograph of constructed area showing setbacks on 4 sides
  - Appendix E-4 (Form for Notice for Occupancy Permit)
- Required (Completion cum Occupancy Part/Full)
- Completion cum Occupancy Date

- User clicks the "Intimate Now" button.

- Building Clerk & Building Inspector receives the alert on SMS, e-mail & dashboard. The Building Clerk verifies the documents & the Building Inspector inspects the site, fills the Completion cum Occupancy Check List and both BC & BI forwards the case to the Final Approving Authority for approval. If the Architect has complied by the norms, Final Approving Authority issues the certificate for Occupancy.

NOTE:

- If any previous forms aren't filled, system must redirect applicant to that form first. Deemed Occupancy permission is granted if no action is taken within 1 month from the date of application. The Certificate format is displayed in Annexure-IV.
- Applicant can view remarks and the application flow in the user interface.

Completion cum Occupancy Inspection List

File number: Architect: File received:

Site Visit done date:

1. Is construction done as per approved drawing? Y/N
2. Construction done as per approved drawing- If YES
   a. Is Drinking water line available? Y/N
   b. Is electricity connection available? Y/N
   c. Is sewage and drainage connection proper? Y/N
   d. Is building material removed from site? Y/N
   e. Is site clean, neatly maintained and no seepage or leakage in walls? Y/N

Construction done as per approved drawing- If NO

a. Is any violation done at constructed building, can it be rectified? If it can't be rectified then municipal engineer shall issue certificate after demolition of violated area or reconstruction – Y/N
   *In case of yes, box will come up for Building inspector to list the MOS violations
   Front [ ] Rear [ ] Side1 [ ] Side2 [ ]
b. Is MOS violated at constructed site? Y/N
c. If MOS violated is compoundable? If it is not compoundable then only after demolition municipal engineer shall issue certificate.
d. If done construction is for such a purpose which may affect health and convenience of nearby residents according to authority opinion.

3. If the building constructed above municipal sewers, sewage drain or mains or under transmission line? Y/N
4. If the owner has explained all necessary measures to the authority for securing the construction work from unsocial elements? Y/N
5. Distance of nearby police station from constructed building? Y/N
6. If building use has not been clarified by the authority that it would be used for people gathering like cinema, theatre, general worship? Y/N
7. Is there any corporation tax pending on constructed building till date? Y/N
8. If painting work of all internal walls, doors and windows is complete? Y/N
9. If painting work of all external walls, doors and windows is complete? Y/N
10. If provision of lift was proposed in the building whether the lift has been installed? Y/N
11. If all internal and external electrification works are complete? Y/N
12. If all flooring works in the building are complete? Y/N
13. If flooring of parking area has been completed and whether it is in usable condition? Y/N
14. If access to the building has been created connecting the main road and whether it is in usable condition? Y/N
15. If rain water harvesting system has been installed? Y/N
16. If construction of all floors of the building block for which completion cum occupancy certificate is to be issued, is complete? Y/N
17. If parapet of balcony, terrace and railing for staircase has been constructed? Y/N
18. If provisions for fire safety have been incorporated in the building? Y/N

Use Case 27: Citizen Search

Description

- This use case describes the process of Citizen Search by File No, Owner Name, Architect, ULB Name – Zone/Ward/Colony Name/Plot

Actors

- Citizen

Pre-conditions

- Data should be available in BPAS

Primary Business Scenarios

- Citizen types the URL for the BPAS system on the browser. On the Home Page under the Tab of Citizen Services ,the user need to click on the citizen search, after that the pop up window will showing. The user needs to click on the owner search and submit. Citizen selects one of the following Search Criteria & enters the details accordingly:
  - File Number
  - Owner Name
  - Architect Name
  - ULB Name & Zone/Ward/Colony Name/Plot
- Search results are displayed as per the Search Criteria in the below format:

PDF Drawing	Scrutiny Report	Sanction Letter	File No	Proposal Date	Architect	Approval Date	Zone	Ward	Colony	Plot No	Owner Name

- Citizen can view & download the drawing PDF, Scrutiny Report & Sanction Letter, and user click on the file no. And view the complete file status as per the below snapshot.

Use Case 28: Make Payment Online

Description

- This use case describes the process of Make Online Payment facility to the Owner.

Actors

- Owner

Pre-conditions

- Data should be available in BPAS

Primary Business Scenarios

- Owner types the URL for the BPAS system on the browser. On the Home Page, under the Tab of Owner Services, the Owner selects the option of Make Payment Online. The new pop-up windows apperars.
- Owner enters the File No. & Mobile number
  - Owner received OTP and enters.
  - Display the memo associated with the file number.
- Owner selects the Memo & clicks the Pay Online button.
- Owner is redirected to the Payment Gateway & pays the fees (Application/Processing Fee).
- After the successful completion of payment, Owner is redirected and user can view and download the receipt, also the user can get the payment receipt from the Architect/citizen console.

Note: SMS/email alert will be generated on each submission, clearance, return, approval, rejection, payment. For details refer section 8.4.6 (SMS/email notification event)

1.2 Non-Functional Requirement

1.2.1 Security

Security Type	How to implement
Password Policy	It is a way to enforce restrictions on user passwords by defining password policies. When Architect will register him/her self, Password policy will be there. For example: Minimum Password Length Passwords must meet complexity requirements like It contains: uppercase letters, lowercase letters, numbers, and characters.
Session Security	Multiple Session for the same user won't be allowed In application-level session time out will be 20 minutes. User will come out of the application session, if user doesn't make any request after 15 Minutes.
Authentication	Lock a user after 3 failed authentication attempts OTP - one-time passwords ensuring that only properly authenticated users are authorized access to application. 1. Application will provide OTP authentication for Citizens/Owners only. 2. When citizen will select model drawing and submits plan, there will be mandatory field for mobile number & email Id. Digital signature will be embedded during sanction letter generation.
Authorization	Will provide Role based access level security for reports. For example: 1. Division head can view division wise reports. 2. CMO can view respective ULB wise report. 3. Authorized head quarter can view state level reports

Audit trail	Will Capture Inward & Outward time for all user for each and every activity. 1. After plan submission and payment done by Architect "Inward" time will be set for BI, BC & DCR Cell 2. After Clearance from BI, BC & DCR Cell "Inward" time will be set for next level authority (Sub Engineer, CMO)
Robot Attack	Will provide necessary antivirus support For every input form, Captcha will be implemented in application
Secure Socket Layer	For all financial transaction (Fee Payment) HTTPS will be used
Web Service Security	Authentication - Authentication uses a user name and password to authenticate a service client to a secure endpoint. At run time, you user will pass credentials to the Web service. HTTPS - In web service major concern is: SOAP messages are transferred in plain text over the network, so solution for this problem is to use HTTPS (SSL) instead of HTTP, so the communication is encrypted. Message Encryption - All of the body of a SOAP message will be encrypted.

1.2.2 Browser Compatibility

The ABPAS interface will be support all following internet browsers.

Browser	Supported Version
Internet explorer	10 and above
Google chrome	40 and above
Mozilla Firefox	30 and above
Safari	6.1 and above
Edge	All
Opera	20 and above

1.2.3 Accessibility

The application will be hosted in State Data Centre and accessible through Internet.

1.2.4 Backup Data

Three types of backup will be included in backup process

1.2.4.1 File and document backu

1.2.4.2 Database backup

1.2.4.3 Binary backup

The proposed back-up software will support the following features:

1. System can be restored through and other OS .This implies that the Backup will be independent of the Operating System to which the app Device/disk library is connected.

2. Backup software will have capability to backup application platform like Oracle, MS- SQL, DB2, Exchange, Lotus Notes, SAPR3, etc. Runs on Windows OS platform.

3. Will support Disk based Backup, so as to provide High Speed Backup on to Disk, and simultaneously stage it to the Tape Device.

4. The backup software will support backup over SAN and LAN.

5. Proposed backup software would support policy based back up for full, Incremental and differential backup (As per CG SDC policy).

6. Backup Server will support open file backup.

7. In case of Multi drive back up, backup software will offer the functionality to bypass the failed drive.

8. Backup solution will support online reporting

9. All databases should have full tape backups every night. In case of recovery same data will be used.

1.2.5 Unicode Compliant (Multi-Lingual Support)

The application will be Unicode Compliant (support Hindi and English). In application there will a language selection option (For Hindi and English). User interface will be available in both the languages.

1.2.6 SMS/Email Notification Event

SMS/Email Notification Event	Sender	Recipient (user/s)
Architect user registration	BPAS Administrator	Architect, CMO
Architect user approval	BPAS Administrator	Architect, CMO
Architect building plan application submission (if colony existing)	BPAS Administrator	Owner, Architect, BC, BI
Architect building plan application submission (if colony new)	BPAS Administrator	Owner, Architect, Colony Cell, BC, BI
Architect building plan application re-submission	BPAS Administrator	Owner, Architect, concerned official (BC/BI/L1 to L2/CMO) who has returned the file to architect
Model drawing submission by citizen (with specific email and mobile)	BPAS Administrator	Citizen, BC, BI
Plan clearance from BC	BPAS Administrator	Owner, Architect, BC, official (L1)
Plan return from BC	BPAS Administrator	BC, concerned person whom BC is returning (Owner/Architect/BI/L1 to L2/CMO)
Plan clearance from BI	BPAS Administrator	Owner, Architect, BI, official (L1)
Plan return from BI	BPAS Administrator	Owner, Architect, BI, concerned person whom BI is returning (Architect/BC/L1 to L2/CMO)
Plan clearance from official in ULB (L1 to L2)	BPAS Administrator	Owner, Architect, official who is clearing, official to whom file is forwarded
Plan return from official in ULB (L1 to L2)	BPAS Administrator	Owner, Architect, official who is returning, concerned person (Architect/BC/BI/L1 to L2) to whom the file/application will go
Plan reject from official in ULB (L1 to L2)	BPAS Administrator	Owner, Architect, L1, L2, BC & BI.
Plan approval from Approving Authority in ULB (L2)	BPAS Administrator	Owner, Architect, Approving Authority (L2)
Plan reject from Approving Authority in ULB (L2)	BPAS Administrator	Owner, Architect, Approving Authority (L2)
Demand note generation	BPAS Administrator	Owner, Architect, BC, Approving Authority
Payment notification for demand note	BPAS Administrator	Owner, Architect, BC
Sanction Letter generation	BPAS Administrator	Owner, Architect, Approving Authority

Plan clearance from colony cell	BPAS Administrator	Owner, Architect, colony cell
Building Commencement Notice	BPAS Administrator	Owner, Architect, BI
Plinth Level Inspection Request	BPAS Administrator	Owner, Architect, BI, BC
Plinth Level Inspection Clearance/Return/Reject by BC	BPAS Administrator	Owner, Architect, BC, Approving Authority (only for clearance)
Plinth Level Inspection Clearance/Return/Reject by BI	BPAS Administrator	Owner, Architect, BI, BC, Approving Authority (only for clearance)
Plinth Level Inspection Approval/Return/Reject by Approving Authority	BPAS Administrator	Owner, Architect, Approving Authority
Plinth Certificate (in case of High Rise)	BPAS Administrator	Owner, Architect, Approving Authority
Service Certificate Request	BPAS Administrator	Owner, Architect, BI, BC
Service Certificate Request Clearance/Return/Reject by BC	BPAS Administrator	Owner, Architect, BC, Approving Authority (only for clearance)
Service Certificate Request Clearance/Return/Reject by BI	BPAS Administrator	Owner, Architect, BI, Approving Authority (only for clearance)
Service Certificate Request Approval/Return/Reject by Approving authority	BPAS Administrator	Owner, Architect, Approving Authority
Service Certificate Generation	BPAS Administrator	Owner, Architect, Approving Authority
Completion cum Occupancy Certificate Request	BPAS Administrator	Owner, Architect, BI, BC
Completion cum Occupancy Certificate Clearance/Return/Reject by BC	BPAS Administrator	Owner, Architect, BC, Approving Authority (only for clearance)
Completion cum Occupancy Certificate Clearance/Return/Reject by BI	BPAS Administrator	Owner, Architect, BI, Approving Authority (only for clearance)
Completion cum Occupancy Certificate Approval/Return/Reject by Approving Authority	BPAS Administrator	Owner, Architect, Approving Authority
Completion cum Occupancy Certificate Generation	BPAS Administrator	Owner, Architect, Approving Authority

Note: For any SMS/email whenever Architect is selected owner will also get the same SMS/email. Only exceptions will be Architect user registration and approval where only Architect will receive.

- Escalation SMS/email in case of pendency (after exceeding the SLA) at BC/BI/DCR will go to BC/BI & CMO immediately.
- Escalation SMS/email in case of pendency (after exceeding the SLA) at BC/BI/DCR for more than 3 days will go to BC/BI & CMO immediately.
- Escalation SMS/email in case of pendency (after exceeding the SLA) at BC/BI/DCR for more than 5 days will go to the BC/BI, CMO and Divisional Joint Director.
- Escalation SMS/email in case of pendency (after exceeding the SLA) at BC/BI/DCR for more than 7 days will go to the BC/BI, CMO, Divisional Joint Director, Deputy Director (SUDA) & Director (SUDA).
- Escalation SMS/email in case of pendency (after exceeding the SLA) at BC/BI/DCR for more than 15 days will go to the BC/BI, CMO, Divisional Joint Director, Deputy Director (SUDA) & Director (SUDA).
- Escalation SMS/email in case of pendency (after exceeding the SLA) at CMO for more than 7 days will go to the CMO, Divisional Joint Director, Deputy Director (SUDA) & Director (SUDA).
- Escalation SMS/email in case of pendency (after exceeding the SLA) at CMO for more than 15 days will go to the CMO, Divisional Joint Director, Deputy Director (SUDA) & Director (SUDA).

1.3 User Access & Accountability Matrix

Designation	Create	Read	Update	Delete	SLA (working days)
1. Architect	1. New Building Plan application	1. Building Plan application (Submitted / Returned/ Approved/ Rejected)	1. Returned Building Plan application as per remarks for specific reason in the application received from Parallel flow/ Officials/ CMO	1. New Building Plan application (in complete before submission)	Architect pays the fees as per demand note generated within 15 days of receiving the intimation otherwise competent authority may reject the application
	2. Work Commencement intimation	2. Status of Submitted Building Plan application (Case details)	2. Plinth Level Completion Certificate request		
	3. Plinth level Inspection intimation	3. Architect Dashboard			
	4. Service Certificate request				
	5. Building Completion cum Occupancy Certificate request	4. Demand Note	3. Service Certificate request		
		5. Sanction Letter	4. Building Completion cum Occupancy Certificate request		
		6. Building Commencement Notice			
		7. Plinth Level Completion Certificate			
		8. Service Certificate			
		9. Building Completion cum Occupancy Certificate			
2. Citizen	1. Model drawing submission	1. Search for submitted application			
	2. Work Commencement intimation	2. Demand Note			
	3. Plinth level Inspection intimation	3. Sanction Letter			
	4. Service Certificate request	4. Building Commencement Notice			
	5. Building Completion cum Occupancy Certificate request	5. Plinth Level Completion Certificate			
		6. Service Certificate			
		7. Building Completion cum Occupancy Certificate			
3. Colony Cell	New colony data	1. Building Plan application (Forward / Return)– Colony data	1. Building Plan application (Forward / Return)–verify Colony data with mandatory remark	NA	3
		2. Colony Cell Dashboard			
4. Scrutiny Engine Scrutiny (Automatic)	NA	1. Building Plan application (Forward / Return)– DCR Drawing part	1. Building Plan application (Forward / Return)– DCR Drawing part with mandatory remark	NA	1. G+1 residential – within 1 working day per corporation
		2. DCR Cell Dashboard			2. G+2 residential – within 2 working days per corporation

					3. G+3 residential – within 3 working days per corporation
					4. Any other – within 5 working days per corporation
5. Building Clerk	1. Demand Note	1. Building Plan application (Forward / Return)–Documents part	1. Building Plan application (Forward / Return)– Documents part with mandatory remark	NA	SLA for work progress notice- 5 days
		2. Building Clerk Dashboard	2. Demand Note		
		3. Building Commencement Notice	3. Plinth level Inspection request		
		4. Plinth Level Completion Certificate request	4. Service Certificate request		
		5. Service Certificate request	5. Completion cum Occupancy request		
		6. Building Completion cum Occupancy Certificate request			
6. Building Inspector	NA	1. Building Plan application (Forward / Return)– Site Plan, Building Drawing part	1. Building Plan application (Forward / Return)– Site Plan, Building Drawing part with mandatory remark	NA	3
		2. Building Inspector Dashboard	2. Demand Note		SLA for work progress notice- 5 days
		3. Building Commencement Notice	3. Plinth level Inspection request		
		4. Plinth Level Completion Certificate request	4. Service Certificate request		
		4. Service Certificate request	5. Completion cum Occupancy request		
		5. Building Completion cum Occupancy Certificate request			
7. Building Officer	NA	1. Building Plan complete application (Forward / Returned)	1. Building Plan complete application (Forward / Returned) with mandatory remark	NA	4
		2. Building Officer Dashboard	2. Plinth level Inspection request		
		3. Building Commencement Notice	3. Service Certificate request		
		4. Plinth Level Completion Certificate request	4. Completion cum Occupancy request		
		4. Service Certificate request			
		5. Building Completion cum Occupancy Certificate request			
8. CMO	Digital Signature of Building permission certificate/ Plinth intimation Certificate/ Service Certificate/ Building Completion cum Occupancy Certificate	1. Building Plan complete application (Forward / Returned)	1. Approve/ Return/ Reject Building Plan application with mandatory remark	NA	4
		2. Status of Submitted Building Plan application (Case details)	**2. Approve / update/modify new Colony data (as per prevailing rule)**		
		3. Demand Note			
		4. CMO Dashboard	3. Plinth level Inspection request		

		5. Dashboard and MIS (ULB Level till individual case)	4. Service Certificate request		
		6. Building Commencement Notice	5. Completion cum Occupancy request		
		7. Plinth Level Completion Certificate request			
		8. Service Certificate request			
		7. Plinth Level Completion Certificate request			
		9. Building Completion cum Occupancy Certificate request			
9. Jt. Director, Divisional Office	NA	1. MIS (Divisional Level till individual case)	NA	NA	NA
10. Dy. Director, SUDA	NA	1. MIS (State Level till individual case)	NA	NA	NA
11. Director, SUDA	NA	1. MIS (State Level till individual case)	NA	NA	NA
12. Commissioner SUDA/ Secretary, Pr. Secretary SUDA/ Chief Secretary	NA	1. MIS (State Level till individual case)	NA	NA	NA
13. Admin user (ULB/Directorate)	New user	1. User details	1. User details	Users (as the case may be)	NA
		2. Access rights	2. Access rights		
14. System Admin	New user/ SMS/ e-mail/ rule	1. User details	1. User SMS/ e-mail/ rule	Users (as the case may be)/ rule	NA
		2. Access rights	2. Access rights		

NOTE: User access can be managed by the system itself. Hence, any change in the user base can be performed any time using the system administration features.

Following levels will be in the workflow in the BPAS software

S. No.	Officers	Level
1	DCR Cell	NA
2	Building Clerk	NA
3	Building Inspector	NA
4	Sub Engineer	1 – Initial Point
5	Assistant Engineer	1
6	Executive Engineer	1
7	Superintendent Engineer	1
8	CMO/Commissioner	2 – Final Authority

Level 1 and Level 2 will be mandatory